



AQ11.3: Activity Questions 3 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

Rank of the Hessian matrix $H_f(x, y, z)$ of the function

$f(x, y, z) = 3x^2 + xz + 2zy - z^2$ at point $(1, 0, 1)$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

1 point

Choose the correct statements about $f(x, y, z) = 3x^2 + xz + 2zy - z^2$ at point $(1, 0, 1)$.

☐

$(0, 0, 0)$ is a critical point of f .

☐

Hessian matrix of f at point (x, y, z) is $\begin{bmatrix} 6x & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -2 \end{bmatrix}$.

☐

Hessian matrix of f at point (x, y, z) is $\begin{bmatrix} 6x & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -2 \end{bmatrix}$.

☐

Rank of the Hessian matrix of f at point $(3, -1, 2)$ is 3.

☐

Rank of the Hessian matrix of f at point $(3, -1, 2)$ is 2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Hessian matrix of f at point (x, y, z) is $\begin{bmatrix} 6x & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & -2 \end{bmatrix}$.

Rank of the Hessian matrix of f at point $(3, -1, 2)$ is 3.

Consider the following function

$$f(x, y, z) = x^3 + xy^2 + x^2 + y^2 + 3z^2$$

Answer questions 3, 4, 5 and 6 based on the function f .

Number of critical points of f is



No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Rank of the Hessian matrix of f at point $(-\frac{1}{3}, 0, 0)$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Rank of the Hessian matrix of f at point $(0, 0, 0)$ is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

Level 2

1 point

Choose the correct statements.

☐

$(-\frac{1}{3}, 0, 0)$ is a saddle point of f .

☐

$(-\frac{2}{3}, 0, 0)$ is a saddle point of f .

☐

$(-\frac{2}{3}, 0, 0)$ is a local minimum of f .

☐

$(0, 0, 0)$ is a local minimum of f .

☐

$(0, 0, 0)$ is a local maximum of f .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-\frac{2}{3}, 0, 0)$ is a saddle point of f .

$(0, 0, 0)$ is a local minimum of f .

Let a denote the shortest distance from the origin to the plane $z = x + y + 3$. The value of a^2 is

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 3

1 point

1 point

Consider a function $f(x, y, z)$. Let v_1, v_2, v_3 be the critical points of the function f and H_k be the Hessian matrix of the function at the point v_k . Then which of the following options is(are) true?

☐

If $H_1 = \begin{bmatrix} -10 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} H_1 = \begin{bmatrix} & & \\ -11 & 10 & 11 \\ 0 & 0 & -11 \end{bmatrix}$, then v_1 is a saddle point.

☐

If $H_2 = \begin{bmatrix} 100 \\ 0 & 20 \\ 0 & 0 & 3 \end{bmatrix} H_2 = \begin{bmatrix} & & \\ 1000 & 2000 & 3 \end{bmatrix}$, then v_2 is a saddle point.

☐

If $H_3 = \begin{bmatrix} 100 \\ 0 & 20 \\ 0 & 0 & 3 \end{bmatrix} H_3 = \begin{bmatrix} & & \\ 1000 & 2000 & 3 \end{bmatrix}$, then v_3 is a point of local maximum.

☐

If $H_1 = \begin{bmatrix} 100 \\ 0 & 20 \\ 0 & 0 & 3 \end{bmatrix} H_1 = \begin{bmatrix} & & \\ 1000 & 2000 & 3 \end{bmatrix}$, then v_1 is a point of local minimum.

☐

If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} H_2 = \begin{bmatrix} & & \\ -1000 & -2000 & -3 \end{bmatrix}$, then v_2 is a point of local maximum.

☐

If $H_3 = \begin{bmatrix} -200 \\ 0 & 20 \\ 0 & 0 & 1 \end{bmatrix} H_3 = \begin{bmatrix} & & \\ -2000 & 2000 & 1 \end{bmatrix}$, then v_3 is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $H_1 = \begin{bmatrix} -10 & 0 \\ 1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix} H_1 = \begin{bmatrix} & & \\ -11 & 10 & 11 \\ 0 & 0 & -11 \end{bmatrix}$, then v_1 is a saddle point.

If $H_1 = \begin{bmatrix} 100 \\ 0 & 20 \\ 0 & 0 & 3 \end{bmatrix} H_1 = \begin{bmatrix} & & \\ 1000 & 2000 & 3 \end{bmatrix}$, then v_1 is a point of local minimum.

If $H_2 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} H_2 = \begin{bmatrix} & & \\ -1000 & -2000 & -3 \end{bmatrix}$, then v_2 is a point of local maximum.

Check Answers

